

# Supply Chain Performance Analysis: A Data-Driven Review

## Final Project Findings & Strategic Recommendations

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# Executive Summary: Three Core Findings from Our Analysis

A comprehensive analysis of our supply chain data has revealed critical insights into performance drivers, operational challenges, and strategic opportunities. The objective was to support data-driven decisions to enhance sales, inventory, and shipping.



## Skincare Dominates the Portfolio

The skincare category is the primary revenue engine, accounting for nearly half of total revenue (£1.1M), indicating its strategic importance.



## Significant Stock Violations Impact Availability

We recorded 98 distinct stock violations, highlighting a critical challenge in inventory management that directly affects product availability and potentially sales.



## Supplier Performance is Highly Variable

A significant trade-off exists between supplier lead times and quality. Our most-used supplier also has one of the highest defect rates, presenting an opportunity for optimisation.



# Our Four-Week Analytical Process Ensured a Rigorous, Structured Approach

01



## Week 1 - Data Foundation: Data Modelling & Preparation

Built a robust data model, then cleaned and preprocessed the dataset to ensure accuracy and consistency. The final dataset was complete with no missing values.

02



## Week 2 - Analysis Phase: Answering Critical Business Questions

Formulated and answered key questions about revenue drivers, supplier performance, and operational efficiency using SQL and Python.

03



## Week 3 - Forecasting Phase: Predicting Future Trends

Developed forecasting models to project future revenue revenue and stock levels, identifying emerging trends trends based on historical data.

04



## Week 4 - Visualisation & Reporting: Delivering Actionable Insights

Consolidated all findings into an interactive dashboard and and this final presentation to facilitate strategic decision-making.



# The Data Cleaning Journey: Preparing Supply Chain Data for Analysis

## Understanding the Dataset



**100 Records,  
28 Columns**

A concise dataset  
containing both numeric  
and categorical  
information for analysis

### A Complete Supply Chain Overview

Contains data on products,  
suppliers, manufacturing, and  
logistics to analyse overall  
efficiency



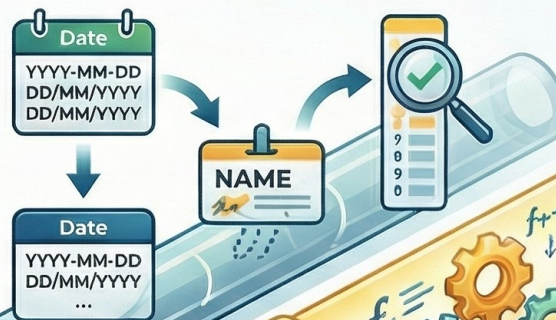
### Initial State: Clean and Complete

The raw data was found to have no  
missing values or duplicate rows.

## The Preprocessing Pathway

### Step 1: Standardise & Validate

Formats for names, numbers, and dates  
were unified and data types were validated



### Step 2: Create Calculated Fields

New fields like "Revenue Difference"  
were created to enable deeper analysis



### Result: An Analysis-Ready Asset

The final dataset is fully clean,  
structured, and suitable for forecasting.







I.

# The Performance Landscape

A comprehensive overview of our supply chain's current state, establishing a baseline with key performance indicators.



# At a Glance: Key Performance Indicators Show Solid Revenue but Highlight Operational Drag



**£2.29M**

Total Revenue Generated



**46K**

Total Products Sold



**17 Days**

Average Supplier Lead Time



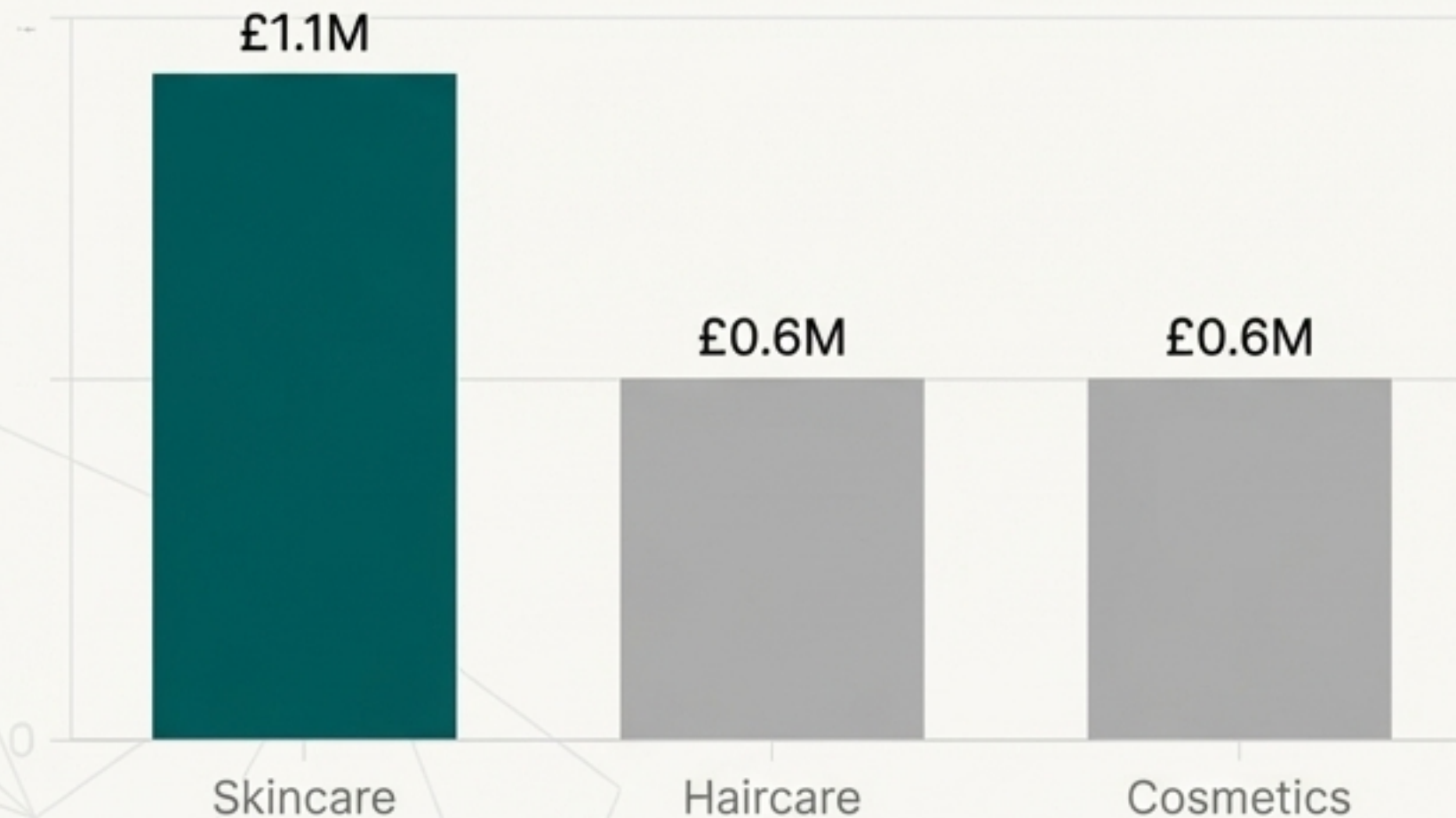
**98 Incidents**

Recorded Stock Violations



# Skincare Products are the Clear Revenue Engine, Contributing £1.1M to the Top Line

## Revenue by Product Category



## Category Contribution

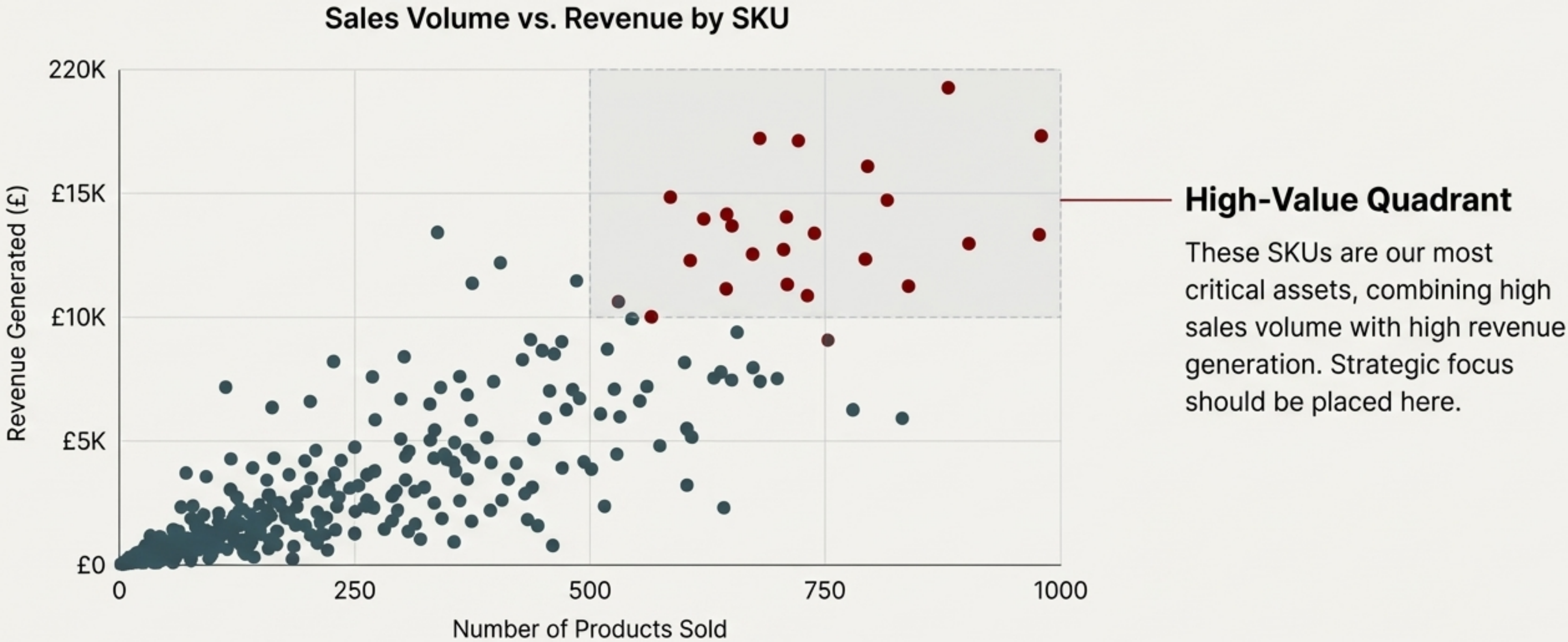
While making up 40% of the product types in the dataset, the skincare category generates nearly half of the total revenue.

Haircare (34% of products) and Cosmetics (26%) contribute almost equally to the remaining revenue.

This concentration of revenue highlights the critical importance of **maintaining supply chain health and product availability** for the skincare line.



# Identifying Our Star Performers: Mapping Sales Volume to Revenue by SKU



*Our data allows us to pinpoint not just popular products, but profitable ones. The next step is to ensure these high-performing SKUs are always available for our customers.*





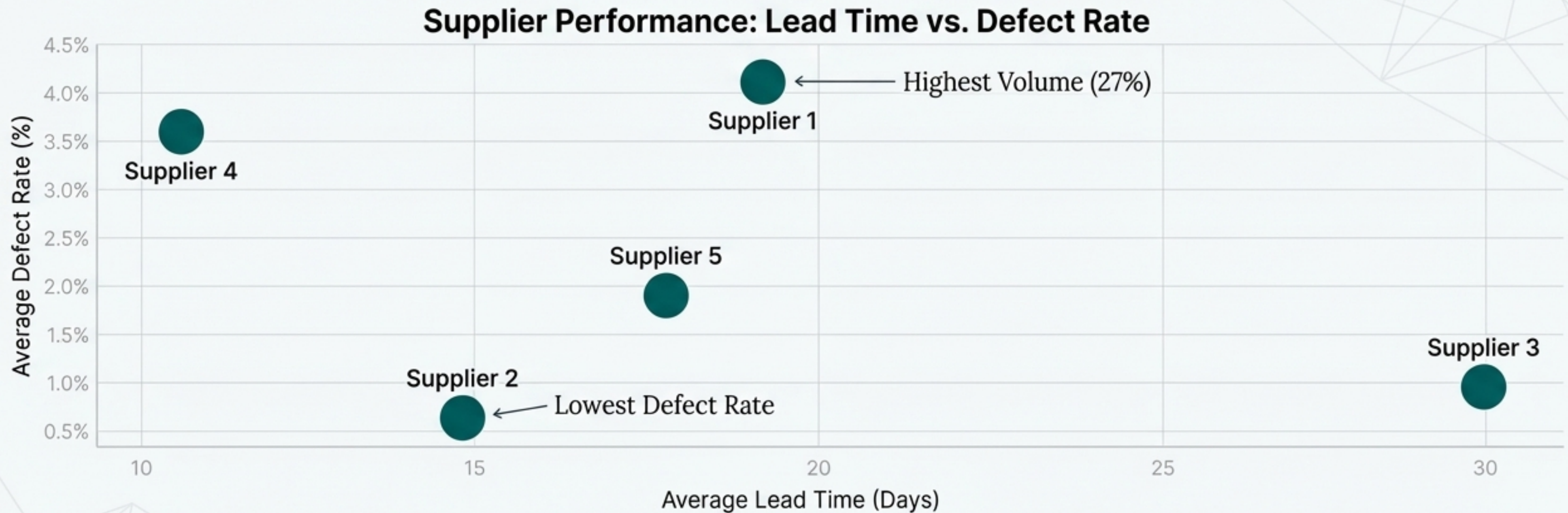
II.

# Partner Ecosystem Analysis

Evaluating the performance and efficiency of our key suppliers and shipping carriers.



# Supplier Performance Reveals a Critical Trade-Off Between Defect Rate and Lead Time



## Our Highest Volume Partner Presents a Quality Risk

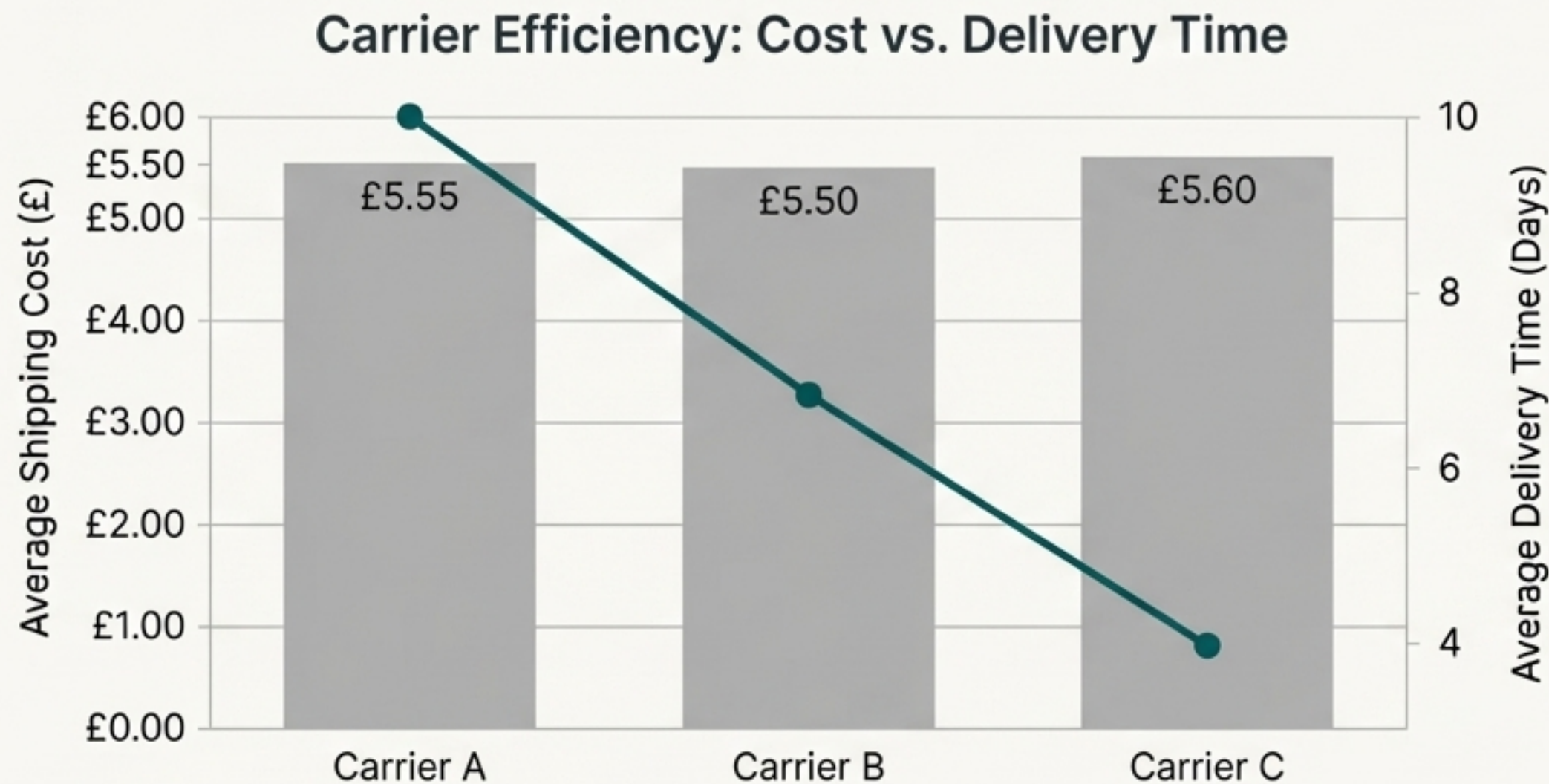
Supplier 1 is our most frequent partner, used in 27% of cases. However, they exhibit one of the highest average defect rates, creating downstream quality control costs and risks.

## Opportunity for Optimisation

Supplier 2 demonstrates the lowest defect rate in the network. Exploring opportunities to shift volume to higher-quality partners like Supplier 2 could significantly improve overall product quality.



# Carrier Efficiency Analysis Shows Delivery Speed is the Key Differentiator, Not Cost



**Performance Profile**

While shipping costs are nearly identical across all carriers, there is a distinct performance gap in delivery times. Carrier B is our most used partner (43% of shipments), offering a moderate delivery speed.

## Strategic Implication

This presents an opportunity to optimise our shipping strategy by allocating shipments based on urgency vs. cost, potentially using the faster Carrier C for priority orders.



The background of the slide features a faint, light gray illustration of several interlocking gears of different sizes, creating a mechanical or industrial theme.

# **III.**

# **Operational Health Check**

Assessing the effectiveness of our internal inventory management and identifying key operational challenges.

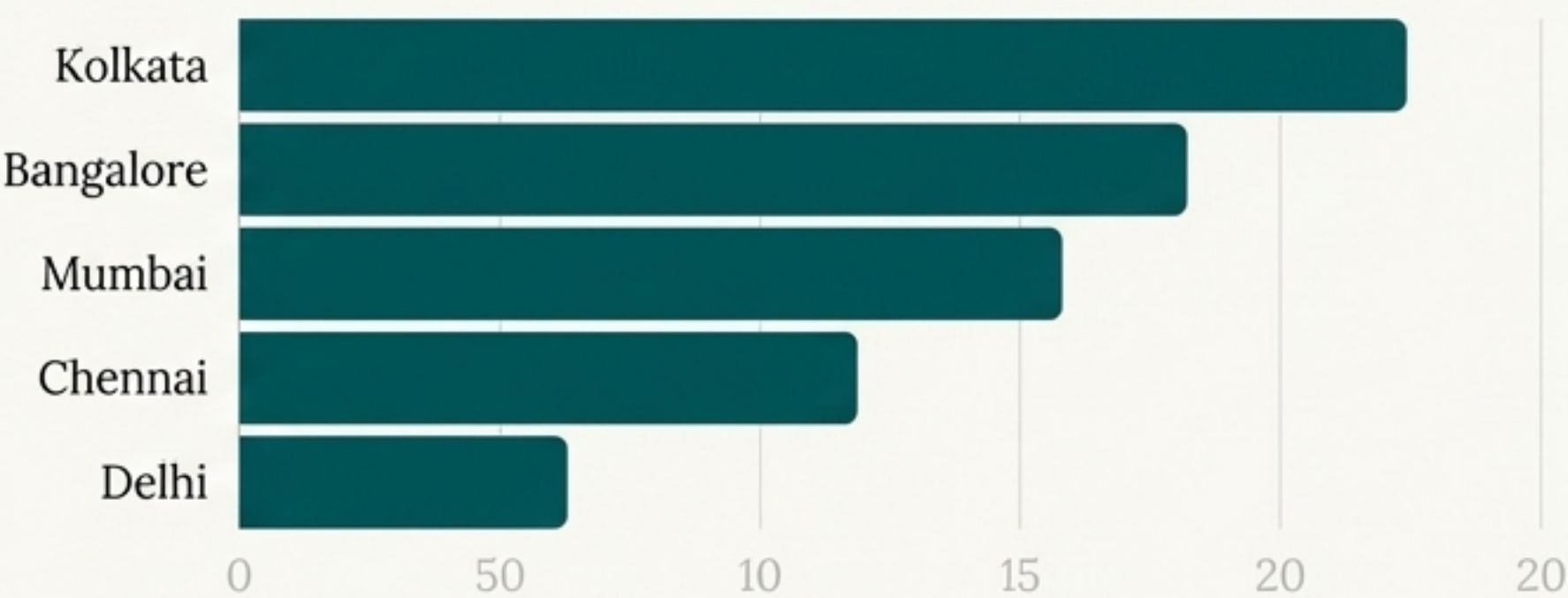


# Inventory Management is a Key Challenge, with 98 Stock Violations Recorded

98

Total Stock Violations

Stock Levels by Location



## The Impact of Violations

These 98 violations represent instances where stock levels fell below required thresholds, leading to potential stock-outs, lost sales, and reduced customer satisfaction. Certain SKUs are repeatedly affected.

The disparity in stock levels across locations, with Delhi holding the least inventory, suggests a need to review our inventory allocation strategy to mitigate violation risks.



The background features a light gray, winding road that recedes into the distance, creating a sense of perspective. Overlaid on this are several thin, light gray geometric lines, including straight lines and arcs, which intersect to form a complex, abstract pattern. The overall aesthetic is clean and modern.

# IV.

## The Road Ahead

Leveraging historical data to forecast future performance and formulate data-driven strategic recommendations.



# Forecasts Project Stable Stock Levels and Fluctuating Short-Term Revenue

Revenue Forecast for Upcoming Periods



\*Note: Represents a periodic forecast, not full annual revenue.\*

Forecasted Stock Levels



The forecast indicates a slight increase in inventory levels, suggesting that stock-outs may not worsen systemically. However, the short-term revenue forecast shows some volatility, reinforcing the need for proactive management of our key product categories.



# Strategic Recommendations to Enhance Performance



## 1. Reinforce the Core Business

Double down on the skincare category. Prioritise its stock allocation, supplier quality, and marketing support to protect and grow our primary revenue stream.



## 2. Optimise the Supplier Mix

Initiate a supplier performance review. Develop a strategy to shift volume from high-defect partners (e.g., Supplier 1) to high-quality partners (e.g., Supplier 2) to reduce defect-related costs.



## 3. Address Stock Violations Head-On

Launch a root-cause analysis for the 98 stock violations, focusing on the most frequently affected SKUs and lowest-stocked locations like Delhi. Re-evaluate safety stock levels.



## 4. Refine Logistics Strategy

Implement a tiered shipping strategy. Use the most cost-effective carriers (e.g., Carrier B) for standard shipments and reserve faster options (e.g., Carrier C) for high-priority or time-sensitive orders.



# Next Steps: Embedding Data-Driven Decision-Making



## Schedule Stakeholder Workshops

We propose a series of deep-dive workshops with relevant teams (Supply Chain, Sales, Operations) to discuss these findings and collaboratively develop detailed action plans for each recommendation.



## Leverage the Live Dashboard

The accompanying Tableau dashboard provides a tool for ongoing, real-time monitoring of these KPIs. We will provide training to ensure it is integrated into regular performance reviews.

This analysis provides a clear roadmap for optimising our supply chain. We look forward to partnering with you to implement these changes.



# Supply Chain Performance: A Data-Driven Overview



Overall Performance Snapshot

**2.29M in  
Total Revenue**

Generated from the sale of over  
46,000 individual products.



**98 Stock Violations  
Recorded**

Highlighting a critical challenge in  
current inventory management.

## Operational Insights & Future Outlook



**Supplier 2 Has  
the Lowest  
Defect Rates**

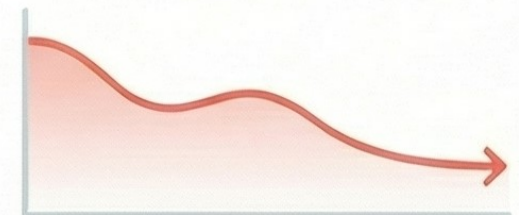
Making them the most  
reliable partner for ensuring  
product quality.



**Carrier B is the  
Most Used and  
Cost-Effective  
Shipper**

Handling 43% of  
shipments while  
maintaining competitive  
shipping costs.

**Revenue is Forecast  
to Dip in 2025**



Projections show a slight decrease from  
2024 levels, requiring strategic focus.

## Revenue by Category

Skincare:  
**1.1M**

Haircare:  
**0.6M**

Cosmetics:  
**0.6M**